



People Analytics

Turnover insights and risk drivers

This presentation summarizes an analysis of employee turnover to identify drivers of attrition and actionable retention strategies.

The study combines *Python*, *SQL*, and Power BI to deliver an interactive dashboard and data-driven recommendations.





Objective

Objective: Analyze employee turnover patterns to pinpoint key drivers and high-risk groups.

Deliverables include a processed dataset for Power BI, statistical analysis of attrition drivers, and recommendations to reduce turnover and improve retention.



Business Questions

Determine overall turnover rate, departments with highest attrition, salary and tenure effects, and job satisfaction impact.

Use findings to prioritize interventions, monitor risk cohorts, and support workforce planning with **actionable insights**.



Tools & Methodology

The dataset was cleaned, encoded, and exported for reporting using *Python (Pandas, Seaborn)* and *SQL*.

Analytical steps included cohort analysis, correlation testing, and visualization in Power BI to produce an interactive HR dashboard and reproducible queries for ongoing monitoring.



Key Metrics

Overall turnover ~16% across the company.

Highest attrition in *Sales* and *Human Resources*. Overtime employees show markedly higher exit rates.





Risk Factors

Primary drivers: *low job satisfaction, frequent overtime, and below-average salary.*

Secondary contributors include short tenure in certain roles and department-specific workload imbalances. Combine factors raise individual risk substantially.



Business Impact

Targeted retention can reduce recruiting and training costs and stabilize critical teams.

Use **risk segmentation** to prioritize interventions, measure satisfaction improvements, and track attrition trends via the Power BI dashboard.



Conclusions

Focus retention efforts on high-risk cohorts identified by satisfaction, overtime, and compensation gaps.

Implement targeted programs, monitor via the dashboard, and iterate with data-driven interventions to lower turnover and maintain workforce continuity.